KENYA MEDICAL TRAINING COLLEGE

FACULTY OF DIAGNOSTIC SCIENCES

DEPARTMENT OF MEDICAL LABORATORY SCIENCES

DIPLOMA IN MEDICAL LABORATORY SCIENCES

YEAR I, SEMESTER I EXAMINATIONS (1.1)

PAPER: LABORATORY MATHEMATICS

DATE: 12/02 /2025	**TIME: 2 HOURS 2.00 – 4.00 PM**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (20 Multiple choice questions)
   - Section 2 (8 short answer questions)
   - Section 3 (2 long answer questions)
2. Attempt ALL Questions
3. Write the COLLEGE NUMBER on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Any examination malpractices will be handled as per the College Examination Policy
6. Ensure you sign the examination register provided

COLLEGE NUMBER .....

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SECTION I: MULTIPLE CHOICE QUESTIONS (20 MARKS)

1. Which of the following units is commonly used to measure blood sugar?
   A. Mg/L
   B. g/L
   C. mMol/l
   D. g/dL
2. Calculate the logarithm of 43:
   A. 1.633
   B. 0.633
   C. 1.778
   D. 0.236
3. What is the molarity of a solution that contains 163,94g Na_3PO_4 in 1 litre of solution?
   A. 2.00M
   B. 1.00M
   C. 1.5M
   D. 2.50M
4. Write in logarithm form $3^5=243$:
   A. $Log_3\ 5=243$
   B. $Log_3\ 243=5$
   C. $Log\ _3 5=729$
   D. $Log\ _3 729=5$
5. Calculte the PH of a base given that its POH value is 10:
   A. 4
   B. 14
   C. 3
   D. 10
6. Which of the following will be value of A, if T=20%?
   A. 0.6655
   B. 0.6667
   C. 0.69897
   D. 0.76669
7. 4.6×10^4 ml is equivalent to how many litres:
   A. 460L
   B. 46L
   C. 4.6L
   D. 0.46L

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8. The energy absorbed and the concentration of solution is a relationship of:
   A. Lambert's law
   B. Beer's law
   C. Daltons law
   D. Beer Lambert's law

9. Moles of solute per litre of solution describes:
   A. Molal
   B. Molarity
   C. Molality
   D. Moles

10. Which of the following determines the strength of a base?
   A. Ka
   B. Kb
   C. Kc
   D. Kw

11. Write 10000 in the form A^n:
   A. 10^4
   B. 10^5
   C. 4^{10}
   D. 4^4

12. The fundamental unit for the length in metric system is:
   A. Centimeter
   B. Millimeter
   C. Inch
   D. Meter

13. $\text{Log}_a M + \text{Log } N$ is:
   A. $\text{Log}_a (M+N)$
   B. $\text{Log}_a (M/N)$
   C. $\text{Log}_a (M \times N)$
   D. $\text{Log}_a M \times \text{Log}_a N$

14 Given a 62% transmittance (T) reading of a given solution, what is the absorbance value?
   A. 2.000
   B. 0.862
   C. 0.174
   D. 1.826

15. Sodium hydroxide is a strong base. What is the pH of 0.02M sodium hydroxide solution?
   A. 12.0
   B. 1.7
   C. 2.0
   D. 12.3

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16. How many milligrams are there in 11 grams?
   A. 110 milligrams
   B. 11,000 milligrams
   C. 110,0000 milligrams
   D. 1,000 milligrams
17. Which of the following criteria cannot be used to choose a primary standard?
   A. Purity
   B. Stability
   C. Solubility in water
   D. Colour formation
18. The negative logarithm to base ten of hydroxyl ion is:
   A. POH
   B. PH
   C. H^+
   D. OH^-
19. Which one is larger quantity, 1000 millimeters or 10 deciliters
   A. 1000 millimeters
   B. 10 deciliters
   C. They are both the same
   D. Cannot be deduced from the data
20. Beer lambert's law gives the relation between which of the following?
   A. Reflected radiation, concentration and light path
   B. Scattered radiation, concentration and light path
   C. Energy absorption, concentration, and light path
   D. Energy absorption, reflected radiation and light path

SECTION 2: SHOET ANSWER QUESTIONS (40 MARKS)

21. Calculate the pH, given the hydroxyl ion concentration (5mks)
   a) POH =8
   b) POH=11
   c) POH=12
   d) POH=9
   e) POH=10
22. Calculate the absorbance, given the following transmittances (5mks)
   a) 70%
   b) 60%
   c) 40%
   d) 30%
   e) 20%

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23. Define the following terms (5mks)

a) Titration
b) Endpoint
c) Indicator
d) Optical density
e) Molarity

24. What is the PH of 0.133 Mol/L HCL solution (5mks)

25. State 5 units used to describe a solute in a solution (5mks)

26. Normal saline is given as 0.85g/100ml. What is its Molarity (5mks)
(Na =23,99, Cl=35.45)

27. Define the following terms (5mks)

a) Stock solution
b) Working solution
c) Ratio
d) Dilution
e) Solutions

28. What is the molarity of a 10% potassium hydroxide (5mks)
(molecular weight of KOH =56)

SECTION 3: LONG ANSWER QUESTIONS (40MARKS)

29. You are provided with concentrated HCL with the following parameters

- Percentage purity 37%
- Molecular weight 36.5 g
- Specific gravity 1.18

Calculate and explain how you can prepare 2 liters whose concentration is 0.2 Molar from the above concentrated acid. (20marks)

30. You are provided with 1 liter of sodium whose concentration is 200mMol/l and 1 liter of potassium whose concentration is 50 mMol/l.
Calculate and explain how you can prepare 1liter of the following combined standards and fill all your values in the table below (20marks).

Concentration of Na+	120	130	140	150
Concentration of K+	3	4	5	6
mL of stock Na+				
mL of stock K+				
mL of diluents(H2O)				
Total volume	1000mL	1000mL	1000mL	1000mL

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